

Here are **150 Python coding questions** focused on improving **logical thinking**, categorized for progressive learning:

-  **Basic & Traditional Programs**
 -  **Strings & Patterns**
 -  **Lists, Tuples & Dictionaries**
 -  **Functions, Recursion & Math**
 -  **Logical Reasoning / Problem Solving**
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PART 1: Basics & Traditional Programs (Q1–Q50)

1. Print the Fibonacci series up to N terms
2. Check if a number is Prime
3. Print all prime numbers in a range
4. Find factorial of a number
5. Check if a number is Armstrong
6. Check if a number is Palindrome
7. Reverse a number
8. Count number of digits in a number
9. Sum of digits in a number
10. Find GCD of two numbers
11. Find LCM of two numbers
12. Swap two numbers (with and without temp)
13. Find power of a number using loop
14. Convert decimal to binary
15. Convert binary to decimal
16. Print multiplication table
17. Sum of first N natural numbers
18. Check if a year is a leap year
19. Check if a number is Strong
20. Check if a number is Perfect
21. Print even numbers up to N
22. Print odd numbers up to N
23. Check if a number is a Harshad number

24. Find sum of squares of digits
 25. Find product of digits of a number
 26. Count number of even and odd digits
 27. Find first and last digit of a number
 28. Check if a number is a Duck number
 29. Check if a number is a Neon number
 30. Convert number to word (123 → One Two Three)
 31. Print all factors of a number
 32. Check if a number is a Kaprekar number
 33. Reverse a number using recursion
 34. Print a number in reverse order
 35. Print number pattern triangle
 36. Print Floyd's triangle
 37. Print Pascal's triangle
 38. Generate AP series
 39. Generate GP series
 40. Find nth Fibonacci number using recursion
 41. Find the sum of an arithmetic series
 42. Print numbers divisible by 3 and 5 in a range
 43. Check if a number is an Automorphic number
 44. Find ASCII value of a character
 45. Print all alphabets from A to Z
 46. Convert Celsius to Fahrenheit
 47. Convert Fahrenheit to Celsius
 48. Solve a quadratic equation
 49. Generate random numbers within a range
 50. Find HCF and LCM using Euclidean algorithm
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PART 2: Strings, Patterns, and ASCII Logic (Q51–Q100)

51. Reverse a string
52. Check if a string is a palindrome

53. Count vowels and consonants in a string
54. Count characters, words, spaces in a string
55. Count frequency of each character in a string
56. Remove all vowels from a string
57. Replace all spaces with hyphens
58. Remove all special characters from a string
59. Capitalize first letter of each word
60. Swap case of all characters
61. Remove duplicate characters
62. Check if two strings are anagrams
63. Find longest word in a sentence
64. Sort words in a string alphabetically
65. Reverse each word in a sentence
66. Convert string to title case
67. Check if a string contains only digits
68. Find first non-repeating character
69. Count number of uppercase and lowercase letters
70. Print ASCII values of each character
71. Encode a string with Caesar cipher
72. Count frequency of words in a paragraph
73. Check if string is Pangram
74. Print pattern of stars in right triangle
75. Print inverted star triangle
76. Print hollow rectangle pattern
77. Print diamond star pattern
78. Print pyramid using numbers
79. Print pattern with characters (A to Z)
80. Create a staircase pattern
81. Print a pattern of alternating 1 and 0
82. Create a hollow triangle
83. Print pattern of Floyd's triangle
84. Alphabetical pyramid (A, AB, ABC...)
85. Print chessboard pattern

86. Pattern of continuous numbers (1, 2, 3...)
 87. Mirror triangle pattern
 88. Hourglass star pattern
 89. Zig-zag pattern with characters
 90. Print multiplication triangle
 91. Create a string without using built-in reverse
 92. Extract numbers from a string
 93. Compare two strings without using ==
 94. Validate password strength
 95. Remove extra spaces from a string
 96. Print string vertically (one character per line)
 97. Print each word of string on new line
 98. Convert string into list of words
 99. Format string using `.format()` method
 100. Replace substring in a given string
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PART 3: Lists, Tuples, Dictionaries, and Sets (Q101–Q125)

101. Find sum of all elements in a list
102. Find max and min in a list
103. Remove duplicates from list
104. Reverse a list manually
105. Count occurrences of an element in a list
106. Find second largest element
107. Merge two lists
108. Find common elements in two lists
109. Split list into even and odd
110. Print elements at even indexes
111. Sum of even numbers in list
112. Multiply all numbers in a list
113. Find average of numbers in list
114. Remove all None values from list

- 115. Sort a list without using `sort()`
 - 116. Create a list of squares of N numbers
 - 117. Tuple unpacking example
 - 118. Check if tuple contains a specific value
 - 119. Convert list to tuple and vice versa
 - 120. Count duplicates in a tuple
 - 121. Iterate over dictionary keys and values
 - 122. Count frequency of words using dictionary
 - 123. Merge two dictionaries
 - 124. Check if key exists in dictionary
 - 125. Remove duplicate values from dictionary
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PART 4: Functions, Recursion, Sorting, and Math (Q126–Q140)

- 126. Create a function to check prime
 - 127. Function to return factorial
 - 128. Recursive Fibonacci function
 - 129. Recursive factorial function
 - 130. Function to find GCD of list of numbers
 - 131. Function to check palindrome
 - 132. Lambda function to square numbers
 - 133. Sort a list of tuples by second element
 - 134. Implement selection sort
 - 135. Implement bubble sort
 - 136. Implement insertion sort
 - 137. Implement linear search
 - 138. Implement binary search
 - 139. Find median of a list
 - 140. Find mode of a list of numbers
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PART 5: Logical Reasoning & Problem Solving (Q141–Q150)

141. Check if two lists are permutations of each other

142. Print all pairs in list with given sum

143. Find missing number in 1 to N list

144. Find duplicate number in list

145. Rotate list elements by K positions

146. Detect if list has a cycle

147. Validate if brackets in string are balanced

148. Generate all subsets of a list

149. Find all prime factors of a number

150. Find max profit from stock prices (buy-sell)